

AMAL P K

Cloud & DevOps Engineer

Kannur, Kerala | amalpk531@gmail.com | amalpk.cloud | LinkedIn | GitHub | +91 9947809632

PROFESSIONAL SUMMARY

Cloud and DevOps Engineer with hands-on experience building and operating cloud-native infrastructure on AWS. Skilled in designing CI/CD pipelines, automating infrastructure with Terraform and Ansible, and deploying containerised applications using Docker and Kubernetes. Experienced in AWS networking, IAM, VPC design, infrastructure automation, and observability practices. Background in backend development with Python and Node.js enables effective ownership across the application lifecycle, from development and deployment to monitoring and operations.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (EC2, EKS, ECS, S3, RDS, ALB, IAM, VPC, ACM)

Containers & Orchestration: Docker, Docker Compose, Kubernetes, Helm, Amazon EKS

CI/CD & Automation: GitHub Actions, Jenkins, Ansible, Terraform, Shell Scripting

Monitoring & Web Servers: CloudWatch, Prometheus, Grafana, Nginx, ALB

Backend & Scripting: Python, Node.js, RESTful API Design

Databases: PostgreSQL, MySQL, MongoDB

Tools & Platforms: Git, GitHub, Linux, Postman

WORK EXPERIENCE

AWS DevOps Engineer Trainee | Dec 2024 – Present

Cokonet Academy, Trivandrum

- Provisioned and maintained cloud infrastructure on AWS (EC2, S3, IAM, VPC, EKS) following security best practices including least-privilege IAM policies and private-subnet isolation.
- Built multi-stage CI/CD pipelines using Jenkins and GitHub Actions, reducing manual deployment effort significantly.
- Containerised applications with Docker and orchestrated deployments on Kubernetes (EKS), ensuring rolling updates and zero-downtime releases.
- Automated infrastructure provisioning with Terraform and Ansible, enabling repeatable, version-controlled environment creation.
- Configured Nginx as a reverse proxy and load balancer for containerised workloads in production environments.

Junior Backend Developer | Jun 2024 – Dec 2024

Adviciya Solutions, Ernakulam

- Designed and developed RESTful APIs in Python, implementing core business logic and optimising PostgreSQL queries for faster response times.
- Integrated GCP Cloud Tasks for distributed background job scheduling and Firebase Cloud Messaging (FCM) for push notifications.
- Managed IAM roles, VPC configurations, and network security rules across cloud environments.
- Collaborated in Git-based workflows with feature branching, code review via pull requests, and structured release management.

Full Stack Web Development Intern | Jan 2024 – Mar 2024

Edunet Foundation (EY GDS & AICTE)

- Completed a 6-week internship focused on MERN stack development and full-stack application architecture.
- Built a personal finance tracker with CRUD operations, JWT/bcrypt authentication, and Chart.js dashboards.
- Developed and integrated RESTful APIs connecting React frontend to Node.js/Express backend.

PROJECTS

Dockerized MERN Application | GitHub

- Containerised a full MERN stack app using multi-stage Docker builds and Docker Compose for single-command spin-up of all services.
- Configured inter-container networking, environment variable injection, and named volumes for persistent MongoDB data.
- Tech: Docker, Docker Compose, MongoDB, Express.js, React, Node.js.

CI/CD Pipeline — GitHub Actions, Docker & EC2 | GitHub

- Built an end-to-end pipeline: builds Docker image on push, runs tests, pushes to Amazon ECR, and deploys to EC2 via SSH.
- Managed secrets via GitHub Actions encrypted secrets and AWS IAM OIDC federation — no long-lived credentials in repo.
- Configured Nginx reverse proxy on EC2 with SSL termination serving the app on a custom domain.
- Tech: GitHub Actions, Docker, Amazon ECR, EC2, Nginx, IAM OIDC.

Multi-Tier AWS Infrastructure with Terraform | GitHub

- Provisioned a production-grade three-tier VPC (public, private, database subnets) across two AZs using reusable Terraform modules.
- Automated creation of ALB, RDS (PostgreSQL), and IAM roles — full stack in minutes.
- Stored Terraform state in S3 backend with DynamoDB locking for safe team collaboration.
- Tech: Terraform, AWS VPC, EC2, ALB, RDS, S3, IAM.

Amazon EKS Deployment | GitHub

- Deployed containerised application on Amazon EKS with node groups, namespaces, Deployments, Services, and Ingress resources.
- Set up AWS Load Balancer Controller and ALB Ingress for external traffic with SSL via ACM.
- Implemented Horizontal Pod Autoscaler (HPA) to scale pods on CPU utilisation for high availability.
- Tech: Amazon EKS, Kubernetes, Helm, ALB Ingress Controller, HPA, ACM, IAM OIDC.

Prometheus & Grafana Monitoring on Kubernetes | GitHub

- Deployed kube-prometheus-stack via Helm on EKS, collecting metrics from all nodes, pods, and control-plane components.
- Built custom Grafana dashboards tracking pod CPU/memory, request latency, and node health with configurable thresholds.
- Configured Prometheus alerting rules and routed critical alerts to Slack via Alertmanager webhooks.
- Tech: Kubernetes (EKS), Helm, Prometheus, Grafana, Alertmanager, Slack.

EDUCATION

B.Tech in Information Technology | APJ Abdul Kalam Technological University, Thalassery | CGPA: 7.5

Diploma in Engineering (Wood and Paper Technology) | Govt Polytechnic College, Kannur | CGPA: 8.6

Higher Secondary Education (Biology Science) | Kerala Board | 70%